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Nonlinear parity-time-symmetric transition in finite-size optical couplers

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Parity-time-symmetric ($\mathcal{PT}$-symmetric) optical waveguide couplers offer a great potential for future applications in integrated optics, such as ultracompact reconfigurable all-optical signal processing. Here, we predict a nonlinearly triggered transition from a full to a broken $\mathcal{PT}$-symmetric regime in finite-size systems described by smooth permittivity profiles and, in particular, in a conventional discrete waveguide directional coupler configuration with a rectangular profile. For these systems, we show that this phase transition occurs in $\mathcal{PT}$-symmetric couplers, regardless of the details of their geometry, therefore suggesting a practical route for experimental realization of such systems. © 2015 Optical Society of America

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One of the fundamental assumptions of quantum mechanics states that operators corresponding to physical observables possess a real spectrum of eigenvalues [1]. The most general class of operators having purely real eigenvalues is pseudo-Hermitian operators [2]. In 1998, Bender *et al.* [3,4] showed that a subclass of pseudo-Hermitian operators, related to the parity-time ($\mathcal{PT}$) symmetry, also possesses a real spectrum. Consequently, Hamiltonians with complex potentials V that fulfill the condition $V(\mathbf{r}) = V^*(-\mathbf{r})$ have real eigenvalues and describe physical phenomena. While the realization of such complex potentials in quantum mechanics is challenging, it is much easier to control them in optics, for instance using the vapor of three-level atoms [5]. Indeed, the light propagation can be described by an optical analogue of the Schrödinger equation, where the complex permittivity $\epsilon(\mathbf{r})$ plays the role of potential with the imaginary part corresponding to gain or loss [6–8].

The optical $\mathcal{PT}$ symmetry was experimentally demonstrated in linear optical waveguide couplers [9], which are important components for future fast, ultracompact, and configurable all-optical signal processing. Optical couplers with gain and loss were also studied in the nonlinear regime [10–13], where unidirectionality [14] and suppression of time reversal [15] were shown. Phenomena related to the nonlinear $\mathcal{PT}$ symmetry were also studied in periodic systems, including solitons [16–23], breathers [24], and their stability [25,26].

Optical $\mathcal{PT}$-symmetric systems, where $\epsilon(\mathbf{r}) = \epsilon^*(-\mathbf{r})$, undergo a transition with the change of the ratio between the imaginary ϵ_{IM} and the real part ϵ_{RE} of the permittivity modulation depth ($\epsilon_{\mathrm{IM}}/\epsilon_{\mathrm{RE}}$). Below a certain threshold value of this ratio, the system is in the full $\mathcal{PT}$-symmetric regime and has purely real eigenvalues [9,27]. For the $\epsilon_{\mathrm{IM}}/\epsilon_{\mathrm{RE}}$ ratio above this threshold, the system is in the broken $\mathcal{PT}$-symmetric regime. In this regime, a pair of modes possesses effective indices that are a pair of complex conjugate numbers, such that one mode experiences gain and the other experiences loss. In linear systems, the transition from the full to the broken $\mathcal{PT}$-symmetric regime (hereafter, simply $\mathcal{PT}$ transition) is usually controlled by varying ϵ_{IM}. Nevertheless, the ratio $\epsilon_{\mathrm{IM}}/\epsilon_{\mathrm{RE}}$ is also influenced by the amplitude of ϵ_{RE} which, in nonlinear systems, can be controlled by varying the incident light intensity. Recently, it has been shown that the nonlinearity can trigger a $\mathcal{PT}$ transition in an infinite periodic array of $\mathcal{PT}$-symmetric waveguides [28]. The dynamics of breathers were also investigated in the finite-size coupler systems with similar geometries to those studied here [29,30].

In this Letter, we show that the nonlinear $\mathcal{PT}$ transition can be observed in ultracompact nonlinear couplers consisting of one or two $\mathcal{PT}$-symmetric waveguides, which can be readily integrated on a chip. We show that the details of the structure geometry do not influence the existence of the $\mathcal{PT}$ transition. This finding is particularly important for practical application, where solid state structures are involved and the index profiles of the waveguides are likely to deviate from exact sine- or cosine-functions often considered in pure theoretical studies. We also find that moving a system further above the $\mathcal{PT}$ transition threshold leads to the blowup of light intensity similar to the one observed in the broken $\mathcal{PT}$ regime. Figure 1 shows the three geometries under investigation: $\mathcal{PT}$-symmetric couplers built of rectangular (a) and sinusoidal (b) waveguides, and a dimer (c), where both gain and loss are located in one waveguiding structure.

To study the nonlinear light propagation in one-dimensional (1D) $\mathcal{PT}$-symmetric structures with Kerr-type nonlinearity, we use the scalar wave equation for the electric field $E(x, z)$:

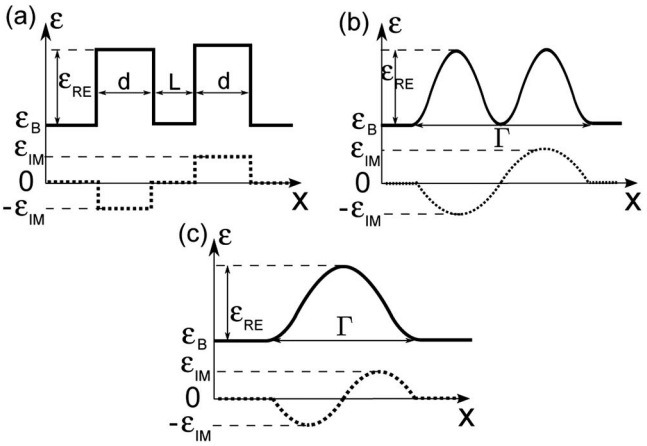

Fig. 1. Geometry of the studied structures with their parameters for different types of $\mathcal{PT}$ symmetric couplers. The size of rectangular waveguides is denoted by d, and the distance between them is L. The size (period) of the cosine-like dimers is denoted by Γ. ϵ_B denotes the background relative permittivity.

$$[\nabla^2 + k_0^2\epsilon(x) + k_0^2\alpha(x)|E|^2]E = 0, \tag{1}$$

where the operator $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial z^2$ denotes the 2D Laplacian operator, $k_0 = 2\pi/\lambda$ is the free-space wave vector, and λ is the free-space wavelength of light. The light propagates along the z-direction, and both the structure and the field distributions are assumed to be invariant along the y-direction. The linear relative complex permittivity distribution along the x-coordinate is described by $\epsilon(x) = \epsilon_B + \Delta\epsilon(x)$, where ϵ_B denotes the background relative permittivity and $\Delta\epsilon(x)$ describes the linear complex permittivity modulation depth. The Kerr nonlinearity strength is quantified by $\alpha(x) = \epsilon_0 \Re e\{\epsilon(x)\} c n_2(x)$, where ϵ_0 denotes the vacuum permittivity, c denotes the speed of light, and n_2 is the nonlinear parameter corresponding to cubic nonlinearity used in the definition of the intensity dependent refractive index $\Delta n_{\mathrm{NL}} = n_2 I$ [31]. The light intensity is related to the electric field in the following way: $I = \epsilon_0 n_B c|E|^2/2$, where $n_B = \sqrt{\epsilon_B}$. The nonlinearity $\alpha(x)$ in the structures studied here is nonzero only inside the waveguides [i.e., if $\epsilon(x) = \epsilon_B$, then $\alpha(x) = 0$].

Using the slowly varying envelope approximation $E(x,z) = \psi(x,z)e^{-ik_0 n_B z}$, Eq. (1) is transformed into the $(1+1)$D nonlinear Schrödinger equation:

$$\frac{\partial\psi}{\partial z} = -\frac{i}{2n_B}\left[\frac{1}{k_0}\frac{\partial^2}{\partial x^2} + k_0(\Delta\epsilon + \alpha|\psi|^2)\right]\psi. \tag{2}$$

While the scalar wave equation [Eq. (1)] is used to obtain the linear modes and dispersion relations for our structures, Eq. (2) is solved using the split-step Fourier method [32,33] to analyze the nonlinear dynamics of the light.

While some of the previous theoretical studies focused on the nonlinear light propagation in an infinite periodic cosine-like $\mathcal{PT}$-symmetric waveguide array [28], here we show that the rich variety of nonlinear phenomena can be obtained in simpler finite-size systems of waveguides that are more feasible from the laboratory viewpoint. First, we study a nonlinear coupler built of two rectangular waveguides presented in Fig. 1(a) and look for the nonlinear transition from the full to the broken $\mathcal{PT}$-symmetric regime. The structure with the following parameters is studied: $d = 1$ μm, $L = 0.1$ μm, $\epsilon_B = 2$, and $\epsilon_{\mathrm{RE}} = 0.02$ at the wavelength $\lambda = 0.63$ μm.

To study the properties of light in our $\mathcal{PT}$-symmetric structures, we solve Eq. (1) using two different methods. First, we study modal properties of the system in the linear regime $[\alpha(x) \equiv 0]$. We look for field profiles that propagate with an effective index n_{eff} without changing their shapes in the form $E(x,z) = \phi(x)e^{-ik_0 n_{\mathrm{eff}} z}$. This allows us to transform Eq. (1) into an eigenvalue problem for the modes of the waveguide described by $\epsilon(x)$ and find the corresponding dispersion relations. We use these dispersion diagrams [shown in Figs. 2(a) and 2(b)] to locate the $\mathcal{PT}$-symmetry breaking threshold.

For the coupler parameters given above, this $\mathcal{PT}$ transition occurs at $\epsilon_{\mathrm{IM}} = 0.0149$. Here, we study the system slightly above the phase transition (at $\epsilon_{\mathrm{IM}} = 0.015$, which corresponds to the broken $\mathcal{PT}$-symmetric regime for low light intensity). The nonlinear dynamics shown in Figs. 2(c) and 2(d) prove

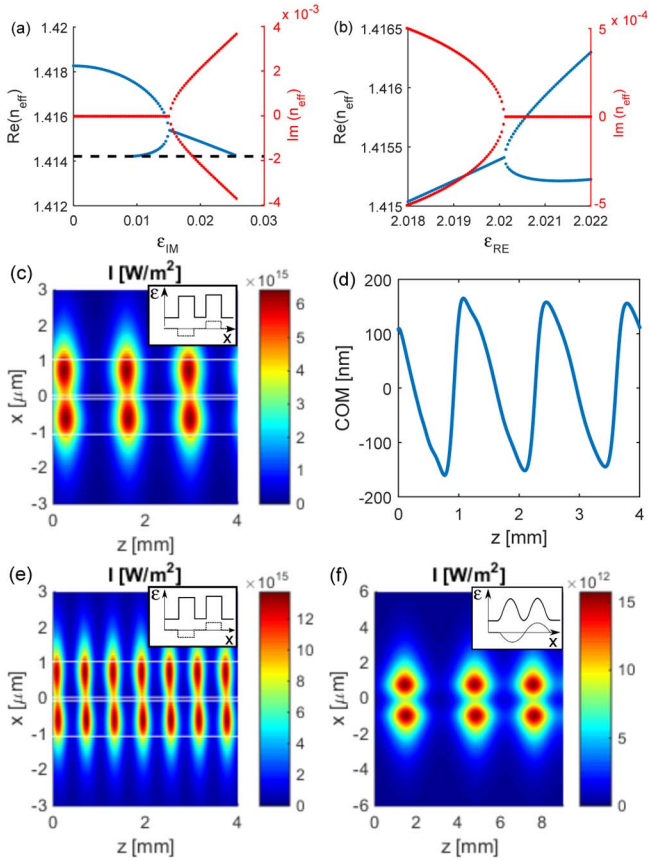

Fig. 2. (a), (b) Linear dispersion relations for a pair of rectangular waveguides shown in Fig. 1(a). The real (blue curves) and imaginary (red curves) parts of the effective index are shown as a function of the (a) imaginary ϵ_{IM} and (b) real ϵ_{RE} part of the permittivity modulation depth. For (a), ϵ_{RE} is equal to 0.02 whereas, for (b), ϵ_{IM} is equal to 0.015. The black dashed line denotes n_B. (c)–(f) Nonlinear dynamics of light propagating in dimers built of (c)–(e) two rectangular waveguides (where $n_2 = 10^{-19}$ m^2/W) and (f) two cosine-like waveguides (where $\alpha = 10^{-19}$ m^2/V^2). (c), (e), and (f) Intensity distribution $I(x,z)$ and (d) evolution of the x-coordinate of the center of mass. The geometry of the waveguides is presented in Figs. 1(a) and 1(b) and in the figure insets. The initial power density P_0 is set to (c), (d) 10^{10} W/m, (e) $3 \cdot 10^{10}$ W/m, and (f) $5 \cdot 10^6$ W/m. White horizontal lines indicate boundaries of rectangular waveguides.

that it is possible to observe the nonlinearly triggered $\mathcal{PT}$ transition in the dimer built of two rectangular waveguides. In this and all the other configurations studied here, we excite the gain mode that initially increases its energy. At the beginning of the propagation, the light intensity is low, and the system is located in the broken $\mathcal{PT}$-symmetric regime. The light is mostly located in the gain waveguide, as it can be seen in Fig. 2(d) showing the x-coordinate of the center of mass (COM) calculated as $\int_{-\infty}^{+\infty} xI(x)\mathrm{d}x / \int_{-\infty}^{+\infty} I(x)\mathrm{d}x$. With the increase of the light intensity, the real part of the permittivity of the waveguides also increases. Since the $\mathcal{PT}$ transition depends on the ratio between the real and imaginary parts of permittivities [28], the increase of ϵ_{RE} drives the system back to the full $\mathcal{PT}$-symmetric regime. (The dispersion relation $n_{\mathrm{eff}}(\epsilon_{\mathrm{RE}})$ presented in Fig. 2(b) shows that with the increase of ϵ_{RE}, the system is transformed from the broken to the full $\mathcal{PT}$-symmetric regime.) After the $\mathcal{PT}$ transition occurs, the light intensity in the gain waveguide reaches its maximum, and then it starts to diminish. At the same time, energy is coupled to the lossy waveguide in which the maximum intensity is reached slightly later and then starts to decay. With the decrease of the total intensity in the coupler, accompanied by the decrease of the nonlinear permittivity modulation depth, the system is transformed back to the broken $\mathcal{PT}$-symmetric regime. This cycle repeats itself during the propagation.

Figure 2(e) presents the light evolution in the same system as the one in Figs. 2(a) and 2(b), but for a higher excitation power density $P_0 = \int_{-\infty}^{+\infty} I(x, z = 0)\mathrm{d}x$. We can observe that with the increase of the initial power, the period of oscillations decreases and the maximum light intensity increases. At the low power level, the oscillation period is equal to ~1.5 mm, and the peak power corresponds to the nonlinear permittivity modulation depth of the order of $1.7 \cdot 10^{-3}$. For high power, the period decreases to 0.6 mm, and the maximum nonlinear permittivity modulation depth increases to $3.5 \cdot 10^{-3}$.

Let us now compare these results for the discrete rectangular waveguide-based coupler with those for the coupler with a smooth cosine-like profile described by

$$\Delta\epsilon(x) = \epsilon_{\mathrm{RE}} \cos^2(2\Omega x) + i\epsilon_{\mathrm{IM}} \sin[2\Omega(x - \pi/4)], \quad (3)$$

and shown in Fig. 1(b). Here, $\Omega = \pi/\Gamma$, Γ denotes the full width of the dimer, and both ϵ_{RE} and ϵ_{IM} are real quantities. While such smooth cosine-like profiles are often used in theoretical studies of periodic structures and waveguides, it may be challenging to realize such profiles in experiments. The cosine-like coupler parameters are $\epsilon_B = 2$, $\epsilon_{\mathrm{RE}} = 0.03$, and $\Gamma = 3$ μm. The imaginary part of the permittivity is chosen to be $\epsilon_{\mathrm{IM}} = 0.0162$, which is just above the $\mathcal{PT}$ transition point located at $\epsilon_{\mathrm{IM}} = 0.016$. It is noteworthy that the dynamics of nonlinear wave propagation in such a structure, shown in Fig. 2(f), resembles that of the discrete waveguide-based coupler [see Fig. 2(e)], suggesting that the nonlinearity-induced $\mathcal{PT}$ transition is not sensitive to the exact refractive index profile of the structure. These results are likely to be important for the experimental observation of the predicted phenomena.

Having established the similarity between light propagation in $\mathcal{PT}$-symmetric structures defined by rectangular and smooth permittivity profiles, we consider the effect of different distributions of both the real and imaginary parts of the dielectric permittivity on the $\mathcal{PT}$ transition dynamics. In particular, we investigate a finite-size $\mathcal{PT}$-symmetric dimer described by

$$\Delta\epsilon(x) = \epsilon_{\mathrm{RE}} \cos^2(\Omega x) + i\epsilon_{\mathrm{IM}} \sin(2\Omega x), \quad (4)$$

with Kerr-type nonlinearity. The geometry of a single dimer described by Eq. (4) is shown in Fig. 1(c), and the parameters are $\epsilon_B = 2$, $\epsilon_{\mathrm{RE}} = 0.05$, and $\Gamma = 3$ μm. Here, the transition from the full to the broken $\mathcal{PT}$-symmetric regime is also observed. The threshold in the linear case is located at $\epsilon_{\mathrm{IM}} = 0.0305$.

The nonlinear propagation of light in a single dimer is presented in Figs. 3(a) and 3(b). It resembles the propagation of light in the unit cell of a periodic array of dimers presented in Figs. 3(d) and 3(f) in [28]. However, in [28], the parameters of the array were chosen in such a way that the energy transfer between the gain and loss regions was visible only from the plot showing the center of mass and not from the color map presenting the intensity evolution during the propagation. In contrast, for the parameters chosen here, this energy transfer, illustrated as the displacement of the center of mass in Fig. 3(b), is also clearly visible from the intensity distribution presented in Fig. 3(a). Indeed, the intensity distribution pattern that changes periodically along the z-coordinate is visibly tilted away from the vertical axis, indicating the energy transfer from the top part of the dimer (gain region) to the bottom part (loss region).

Figures 3(a) and 3(b) show that the system is initially (for low light intensity) in the broken $\mathcal{PT}$-symmetric regime. The gain mode of the system is excited, and the energy of this mode increases. Similar to the case shown in Fig. 2, this increase is accompanied by the increase of the real part of the permittivity due to the Kerr effect. This increase brings the system back to the full $\mathcal{PT}$-symmetric regime. Initially, the energy is stored in the gain mode of the broken $\mathcal{PT}$-symmetric system. In the full

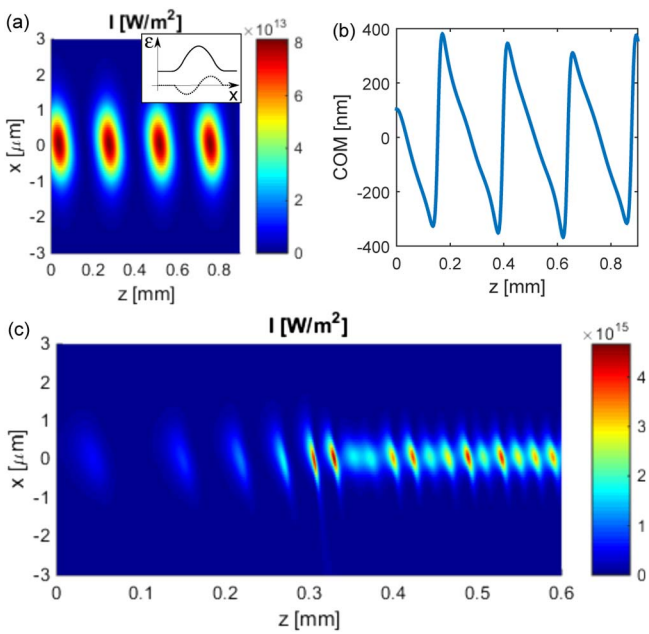

Fig. 3. Nonlinear dynamics of light propagating in a single dimer described by Eq. (4). (a), (c) Intensity distribution $I(x, z)$ and (b) evolution of the x-coordinate of the center of mass corresponding to subplot (a). The geometry of the structure is shown in Fig. 1(c) and in the inset of supblot (a). The parameters are $\epsilon_{\mathrm{RE}} = 0.05$, $\alpha = 10^{-19}$ m²/V², and initial power density $P_0 = 10^8$ W/m; the imaginary part of permittivity ϵ_{IM} equals (a), (b) 0.031 and (c) 0.035.

$\mathcal{PT}$-symmetric regime, this mode is no longer an eigenmode of the system, and the distribution of light (still experiencing gain as it is localized mostly in the gain region) is attracted to the center of the dimer, where the permittivity is the highest [28]. Therefore, the beam possesses a nonzero transverse momentum directed toward negative x-direction. Due to this momentum, the beam crosses the center of the dimer and, as a result, is mostly localized in the region with loss. Consequently, the total energy rapidly decreases, and the system is transformed back to the broken $\mathcal{PT}$-symmetric regime. In this regime, the part of the energy stored in the loss mode vanishes quickly, and the part stored in the gain mode starts to grow again. This cycle repeats, as is seen in Figs. 3(a) and 3(b).

It is noteworthy that, in the case of a single dimer for which the parameters are chosen in such a way that the system is higher than the $\mathcal{PT}$ transition ($\epsilon_{\mathrm{IM}} = 0.035$ while it was 0.031 in the example discussed above), the nonlinear transition also occurs, but it has a different character. The propagation of light in such a system is presented in Fig. 3(c). We see that in the initial phase of the propagation (up to 0.3 mm), the period of oscillations decreases. With the decrease of the period, the energy is transferred from the gain part of the dimer to the lossy part within a shorter distance. At the beginning of the propagation, the transverse component (toward negative x values) of the beam momentum is low enough so that the beam stays confined in the dimer. During the propagation, this momentum increases and, at the propagation distance $x \approx 0.3$ mm, it causes a significant quantity of energy to be outcoupled from the dimer and to propagate toward negative values of x. After this point, the propagation pattern becomes more regular: the period of oscillations is kept constant, and only the peak intensity reached during the oscillations changes. For longer propagation distances, the system relaxes into a stable propagation regime. The irregularity of the propagation in the system located higher above the $\mathcal{PT}$ transition can be explained by the fact that the light intensity here is 20 times larger than that for the system closer to the $\mathcal{PT}$ transition. At the same time, the period of the oscillations is about four times smaller and, consequently, the changes in the light intensity are much more abrupt than in the system closer to the $\mathcal{PT}$ transition. These rapid changes cause more irregular behavior of the system high above the $\mathcal{PT}$ transition. For the systems located even higher above the $\mathcal{PT}$-transition threshold (higher values of ϵ_{IM}), the system slips into the blowup regime and the light intensity increases to the value orders of magnitude higher than the damage threshold of typical materials used in integrated optics.

In conclusion, we have studied nonlinear dynamics of light propagating in finite-size parity-time-symmetric optical couplers. We have shown that the nonlinearly triggered transition from the full to the broken $\mathcal{PT}$-symmetric regime can be observed for various finite-size couplers built of two rectangular and cosine-like waveguides. We have shown that the nonlinearly triggered transition between the full and the broken $\mathcal{PT}$-symmetric regime also occurs in a single $\mathcal{PT}$-symmetric cosine-like dimer. Moreover, we have found that the light propagation in smooth and discrete waveguide couplers is qualitatively similar. These results suggest a practical and simple route to experimental verification and applications of this type of nonlinear $\mathcal{PT}$ transition.

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